

JAYANTH DASAMANTHARAO

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PROFESSIONAL EXPERIENCE

AI Engineer - Fedway Associates, EliteUS Software Solutions Aug 2024 - Present | Edison, NJ

- Designed and deployed agentic AI platforms using AutoGen, LangGraph, and custom intent-routing orchestration to dynamically route user queries across multiple secure AI agents based on role and context.
- Built memory architectures combining session persistence, summarization pipelines, Parquet storage, and vector databases (Chroma, Pinecone, FAISS) to support long-running AI workflows and RAG.
- Developed semantic retrieval systems with recursive chunking, hybrid search, and LLM-based reranking, indexing 10,000+ pages and serving 12,500+ users with high relevance and low latency.
- Built developer facing AI services and APIs using Python, FastAPI, Node.js, and cloud infrastructure (AWS, Azure), integrating authentication, guardrails, and production monitoring into live customer portals.

Data Science Intern, Harvest Software Solutions June 2023 - Sept 2023 | Jacksonville, Florida

- Engineered scalable ML pipelines and monitoring systems for large advertising datasets, increasing campaign visibility from 15% to 45% while ensuring reliability and performance.
- Built ML services and NLP APIs using XGBoost, H2O AutoML, and FastAPI, improving sentiment classification accuracy by 65% and campaign spend efficiency by 50%.
- Designed SQL/Python data processing and evaluation pipelines, implementing hash maps, sliding windows, priority queues, batching strategies to aggregate, validate, analyze data, improving 40% engagement quality.

Application Development Associate, Accenture June 2021 - Aug 2022 | Hyderabad, India

- Developed and optimized Python-based ETL and data processing pipelines using hash maps, sets, indexed lookups to efficiently clean, deduplicate, and transform large datasets, reducing production issues by 50%.
- Implemented SQL workflows and joins with partitioning, indexing, and window functions, enabling efficient aggregation and analysis over millions of records in AWS Athena and Redshift.
- Applied batching, caching, and algorithmic optimizations for data validation and pipeline execution, improving system performance and reliability by 25%.

TECHNICAL SKILLS

Programming:	Python, R, SQL. Frameworks: PyTorch, TensorFlow, Hugging Face Transformers, FastAPI.
Machine Learning :	Model Deployment, Evaluation, Monitoring, Debugging, Feature Engineering.
GenAI & LLMs:	LLM fine-tuning, RAG, Prompt Engineering, LangChain, Vector Databases, Multimodal AI.
AI Platforms/Agents:	Agentic systems, orchestration, memory architectures, tool calling, guardrails.
APIs & Services:	REST APIs, API versioning, authentication, developer tooling.
Cloud & MLOps:	PostgreSQL, Data Warehousing, AWS (Bedrock, Athena, Redshift), GCP, Azure, Vectordb.

EDUCATION

Rutgers University, Master of Science in Data Science Sept 2022 - May 2024 | New Brunswick, NJ

Personal Website: <https://jayanthdasamantharao.github.io/>

Andhra University B.Tech in Electrical and Electronics Engineering 2017 - 2021 | Visakhapatnam, India

RESEARCH PROJECTS

Self-Driving Cars (Computer Vision): Built autonomous driving cars with vision pipeline using CNNs, YOLO, Faster R-CNN, achieving 95%+ object detection accuracy.

Multimodal GenAI Chatbot (Production): LLM-powered chatbot with RAG, embeddings, caching, and text + image outputs, deployed on cloud APIs along with Human-In-The-Loop architecture and optimization.

Bayesian Classification: Implemented Bayesian logistic regression in PyMC3 to predict diabetes status, incorporating informative priors, achieving 0.85 F1-score.

For additional projects and code samples: <https://github.com/JayanthDasamantharao>